Single-Band Stripe Noise Removal in Multispectral Remote Sensing Images Based on Semisupervised Disentangled Transformation Network

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***Abstract*— The presence of stripe noise in multispectral data is a common issue caused by various factors during the imaging process. This noise severely degrades image quality and imposes limitations on downstream tasks. Although deep learning-based methods have demonstrated promising results in destriping, they often encounter challenges due to the disparity between the stripe noise distribution in simulated and real images. As a result, their destriping performance on real data is significantly hindered. To address this challenge, we propose a semisupervised disentangled transformation network (SDTNet) that encourages the model to learn the real stripe noise distribution through image decoupling and noise transformation using simulated and real data. SDTNet consists of the simulated and real image branches, which are subject to supervised and unsupervised constraints, respectively. They are jointly trained to enhance each other mutually. Furthermore, we introduce a decoupling strategy that effectively preserves the clean background component using self-consistency and adversarial losses. Instead of directly converting the whole image from the simulated domain to the real domain, SDTNet focuses on the relatively simpler task of converting the stripe noise component while maintaining the consistency of the image background. Extensive experimental evaluations on various datasets demonstrate the superior destriping performance of the proposed SDTNet compared to other methods, particularly in effectively removing stripe noise from real images.**

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***Index Terms*— Multispectral remote sensing image, semisupervised disentangled transformation network (SDTNet), stripe noise removal.**

I. Introduction

MULTISPECTRAL remote sensing images possess a wealth of spatial and spectral information, rendering them highly valuable resources. However, the unstable imaging system, influenced by various factors such as stray light interference, slit pollution, and instrument instability, can lead to variations in the digital numbers obtained from adjacent detectors, even when capturing the same ground material. The resulting stripe noise in the acquired data due to this inconsistency significantly degrades the data quality, thereby undermining the effectiveness of subsequent tasks [1], [2], [3], [4], [5]. Consequently, stripe removal assumes a crucial role in the remote sensing field, enhancing data quality and facilitating subsequent implementations' advancement.

Various approaches have been developed to address the problem of this interference problem in multispectral remote sensing images. These approaches can be broadly categorized into four groups: statistical matching-based methods [6], [7], filtering-based techniques [8], [9], [10], optimization-based methods [11], [12], [13], and deep learning-based approaches [14], [15]. Nevertheless, traditional destriping methods often face limitations, including poor restoration performance when dealing with complex ground materials, the inability to handle images with prominent characteristics in the transformed domain, and the reliance on rigid assumptions that can invalidate the effectiveness of stripe removal. In recent years, deep learning-based techniques have emerged as a promising solution [16], harnessing the representation capabilities of convolutional neural networks (CNNs) in the remote sensing field [17], [18], [19], [20], [21], [22]. Subsequently, significant advancements have been made in remote sensing image destriping with the aid of deep learning techniques.

Among the existing destriping methods, deep learning-based methods have emerged as the optimal approach for achieving superior destriping performance [23]. There are generally two types of methods: self-supervised and supervised learning. The former learns meaningful representations or features from unlabeled data without the requirement of external labels [24], [25], [26]. These methods can better utilize the

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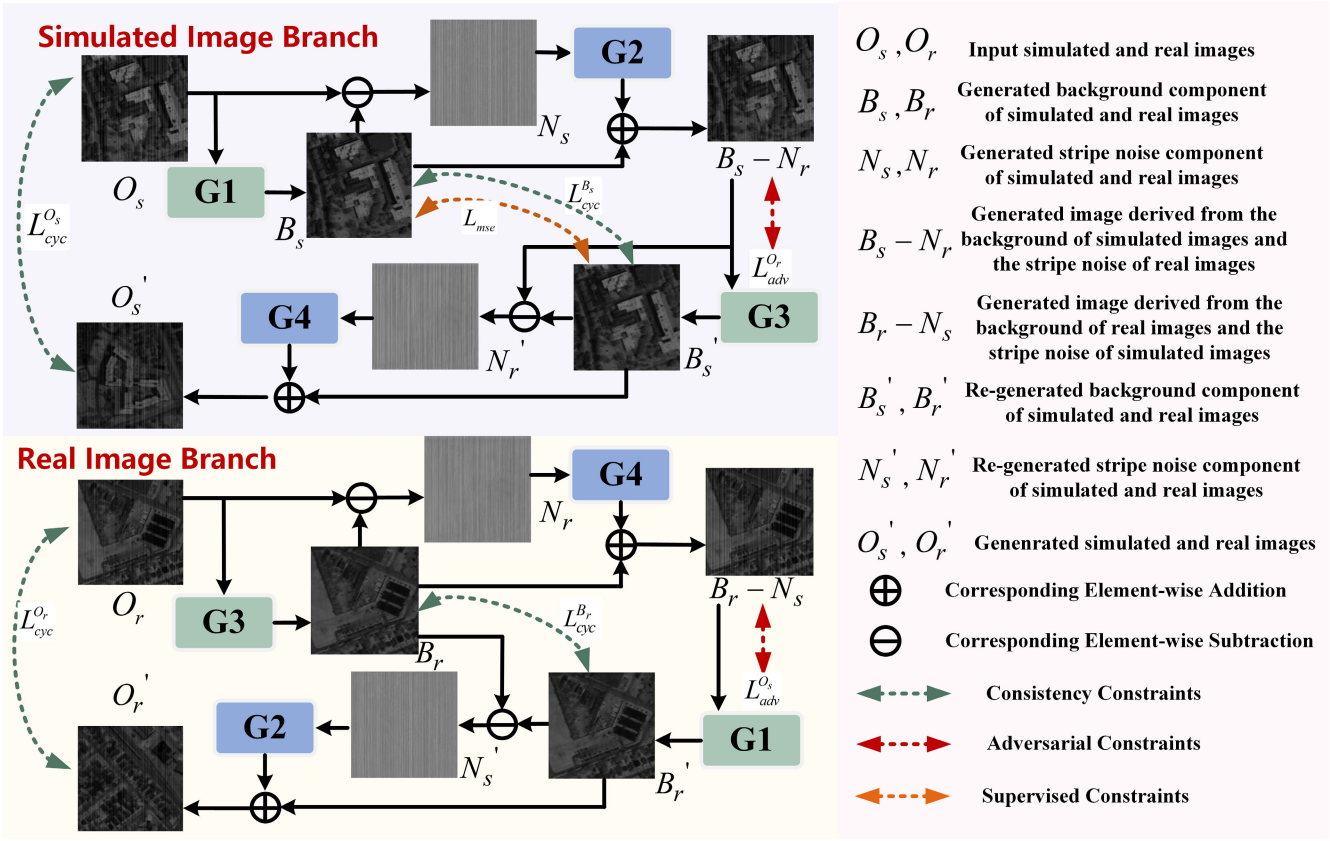


Fig. 1. Structure of the proposed SDTNet. The proposed method comprises two branches, namely the simulated image branch and the real image branch, intending to bridge the gap between the two domains. Rather than relying on direct conversion, we integrate stripe noise removal and generation within each module, thereby enhancing the overall robustness of the method. Moreover, we introduce self-consistency loss and adversarial loss to better preserve the identity of the clean background component.

latent information of the data, thus learning representations or features with greater generalization ability. However, the design of the self-supervised task and the higher computational resource requirements limit the performance of self-supervised learning methods. The supervised models leverage the robust feature extraction capabilities of the network, enabling efficient learning of the mapping relationship between input and desired output images, thereby generating high-quality clean images. Currently, most deep learning-based methods focus on enhancing destriping performance by incorporating various modules to improve the network's learning capacity [27], [28], [29]. However, these models trained on simulated samples, generated from manually designed degradation models, often exhibit poor performance when applied to real test data. This discrepancy arises because these simulated samples do not precisely replicate the actual degradation process. Moreover, the complexity of spectral imaging systems and the multitude of interfering factors during imaging make it difficult to create an accurate mathematical model for the degradation process of multispectral images [30]. Thus, developing an unsupervised method for the deep-learning models to learn the noise distribution of real data offers an effective solution to this challenge.

We introduce a semisupervised disentangled transformation network (SDTNet) aimed at encouraging the model to learn the real image noise distribution through image decoupling and noise transformation. This approach is based on the observation that destriping involves separating the input degraded image into two distinct components: the clean background component and the stripe noise component. As illustrated in Fig. 1, SDTNet comprises two main branches: the simulated and real image branches. These branches are designed for decoupling and generating the simulated and the real images, respectively. During the training process, the simulated image branch is guided by supervised constraints using the ground truth maps, while unsupervised constraints guide the real image branch through the discriminative model. This enables the network to acquire information regarding the distribution of noise in real images. By decoupling and transforming the simulated and real images, SDTNet preserves the clean background component while converting the stripe noise component between the two branches. SDTNet stands out from existing deep learning methods by being designed as a comprehensive end-to-end learning framework that seamlessly combines the generation and removal of stripe noise images. This design choice overcomes the challenge of explicitly designing complex stripe degradation models for real images. Instead, SDTNet learns the stripe distributions directly from real degraded images affected by noise. These real-world noise distributions are much closer to the actual noise patterns encountered in practice, in contrast to artificially designed degradation models.

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To sum up, the main contributions can be outlined as follows.

1) In this study, we propose a novel multispectral remote sensing image destriping algorithm named SDTNet, which focuses on motivating the model to learn the real stripe noise distribution by employing image decoupling and noise transformation. In contrast to using a hand-crafted stripe noise generation model, learning the physical degradation directly from real images provides a more accurate approximation of the real data implicitly.
2) We present a decoupling strategy that effectively preserves the consistent background by incorporating self-consistency loss and adversarial loss. Unlike directly converting simulated images to real images, our proposed method specifically targets the conversion of the stripe noise component, while discarding the clean background component. This approach significantly reduces the disparity between simulated and real images, thus bridging the gap between the two domains.
3) We developed strip removal and noise conversion generators tailored for the simulated and real images, respectively. These generators are interactively employed in both branches, enabling mutual enhancement of model performance between the simulated and real image branches.

II. Related Works

A. Traditional Multispectral Remote Sensing Destriping Method

Traditional destriping methods can generally be classified into three categories: statistical matching-based, filtering-based, and optimization-based approaches. Statistical methods operate under the assumption that each column of pixels in the image exhibits consistent statistical characteristics when the imaging system captures ample historical data. Based on this observation, statistical matching-based methods aim to remove stripes by adjusting the data within each column based on statistical calculations. Representatives of statistical matching-based methods include histogram matching and moment matching (MM). For instance, Horn and Woodham [6] employed histogram adjustment of each column to a clean reference for Landsat images destriping. However, when the available data are insufficient, the statistics information of different columns can vary, leading to inferior destriping performance. To address this limitation, Wegener [7] and Gadallah et al. [31] developed alternative techniques in the methods as mentioned above. There are also alternative approaches. For instance, one method involves the maximum a posteriori estimation of gain and offset parameters for each detector based on a given observed image [32]. Another method addresses stripe noise by performing radiometric normalization of different columns in the image [33].

Filtering-based destriping methods aim to leverage the improved visibility of various image components in the other domain compared to the original domain. These methods involve transforming the original data into the transformed domain to remove the noise signal. In cases where the noise signal exhibits strong characteristics in the spatial domain, the Fourier transform methods can be employed to design suitable filters in the frequency domain for destriping [8]. In contrast, wavelet-based methods learn image features through fundamental wavelet functions while taking into account the directional characteristics of stripes in remote sensing images [34]. Münch et al. [35] introduced the Fourier transform methods that integrate wavelet decomposition, effectively harnessing the scaling and directional properties to enhance the performance of destriping. In addition, several destriping methods have been proposed that combine statistical matching and filtering techniques [9], [10], [33], [36], aiming to fuse the strengths of both approaches to achieve improved destriping outcomes.

Optimization-based methods generally involve two crucial steps: objective function design and stripe removal. These methods aim to estimate the clean image from its striped counterpart by optimizing the designed objective function, which typically comprises fidelity and constraint terms. Constraint terms can include total variation (TV) [11], [37], low-rank property [38], [39], or sparsity [40], [41]. Optimization-based methods can be categorized based on the data processing approach, which includes variation, low-rank matrix, and low-rank tensor methods. Variation-based methods generate a clean image by considering prior knowledge and utilizing a variation model [11], [12], [37]. Another type of method assumes that the data exhibit a low-rank characteristic. These methods can be further divided into matrix-based approaches [13], [42] and tensor-based methods [43], [44] based on the form of data processing. Additionally, there exists a method that optimizes destriping performance by combining both approaches [45]. Furthermore, tensor decomposition effectively reduces the dimensionality of data while preserving the important information of the original data by decomposing high-dimensional data into low-rank tensor forms. It is widely used in tasks such as data dimensionality reduction, feature extraction, and image restoration in the hyperspectral image analysis field.

Each of the above methods possesses its unique advantages as well as limitations. While statistical matching-based destriping methods are relatively efficient in terms of implementation, their effectiveness can be limited in practice due to the strong assumption they rely on. This assumption may not hold in the presence of complex ground object distributions, which can considerably impact the performance of these methods in destriping tasks. Additionally, in certain cases, such as when dealing with broken stripes that only partially cover a column, these methods tend to produce unsatisfactory results due to incorrect matching information. On the other hand, filtering-based methods generally achieve satisfactory destriping performance when dealing with stripes that exhibit strong periodicity. However, their effectiveness can be limited when confronted with real stripes that possess complex characteristics. Although optimization-based methods demonstrate superior performance, they often suffer from computationally intensive processes, leading to slower processing speeds, particularly when handling large datasets. As a result, these methods are not suitable for applications requiring real-time performance.

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B. Deep Learning-Based Multispectral Remote Sensing Destriping Method

In recent times, deep CNN-based models have gained significant traction for image destriping tasks, drawing inspiration from the remarkable achievements of deep learning across diverse domains, such as multihybrid attention mechanisms [46], spatial–spectral information interaction [47], and multiresolution feature guidance [48]. These deep CNN-based models have emerged as the most advanced methods for destriping, as they eliminate the need for handcrafted assumptions and priors [14], [15]. Supervised CNNs exhibit strong feature extraction capabilities, enabling them to effectively address arbitrarily mixed stripes. Subsequently, several destriping methods have been introduced to enhance restoration performance from different perspectives, such as residual learning strategy [27], multiscale features [28], directional characteristics of stripes [29], spectral and spatial intrinsic features [49], as well as incorporating local and global image information [50]. These approaches aim to strengthen destriping performance by leveraging specific features or modules and incorporating them into the deep CNN models.

III. Methodology

A. Overview

Our objective is to estimate the clean background $\boldsymbol{B}$ from a real degraded image $\boldsymbol{O}$. However, the notable disparity between the simulated training image $\boldsymbol{O_s}$ and the real test image $\boldsymbol{O_r}$ can significantly hamper the performance of fully supervised learning-based methods. To overcome this challenge, we introduce SDTNet, which effectively approximates the distribution of stripe noise in real data through implicit learning, as illustrated in Fig. 1. In SDTNet, two branches are employed to decouple images and transfer noise for simulated and real data, respectively.

Specifically, we employ four generators (G_1, G_2, G_3, and G_4) and two discriminators (D_1 and D_2) in SDTNet. G_1 and G_3 are utilized to decouple the clean background component and further extract the stripe noise component from simulated and real images, respectively. Additionally, G_2 and G_4 are responsible for transferring the stripe noise, with G_2 generating the stripe noise from the simulated domain to the real domain, and G_4 performing the reverse process. The discriminator D_1 is trained to differentiate between true and false instances of $\boldsymbol{O_s}$ and $\boldsymbol{B_r} - \boldsymbol{N_s}$, thereby encouraging the generative model G_2 to reduce the disparity in noise distributions between $\boldsymbol{B_r} - \boldsymbol{N_s}$ and $\boldsymbol{O_s}$. Similarly, the discriminator D_2 is employed to enhance G_4 in transferring the stripe noise from the real domain to the simulated domain. As a result, G_1 and G_3 can primarily concentrate on the removal of simulated and real stripe noise, respectively.

In general, this design offers two key benefits. First, the proposed SDTNet establishes a robust relationship between real and simulated degraded images by iteratively decoupling and generating the stripe noise component. This strengthens the connection between the two image domains. Second, the overall architecture facilitates feature propagation in both tasks, resulting in a mutually reinforcing effect that enhances the performance of the model.

B. Loss Function

In this section, we provide a comprehensive explanation of the loss functions employed in our approach. To facilitate the removal of stripe noise and generate high-quality images, we employ a combination of unsupervised adversarial loss, self-consistency loss, and supervised mean squared error (MSE) loss at each intermediate stage. Consistency constraints are applied to ensure the coherence of information between the background images and to maintain consistency between the input images and the final generated images.

1) Adversarial Loss: For the input simulated and real images, making them adversarial constraints against the images $\boldsymbol{B_s} - \boldsymbol{N_r}$ and $\boldsymbol{B_r} - \boldsymbol{N_s}$ generated in the first stage, respectively, can be achieved in an unsupervised manner to facilitate the transformation generation of striped noise images as follows:

$$\begin{aligned} L_{\text{adv}}^{O_s} &= \mathrm{E}_{O_s}\left[\log D_1(\boldsymbol{O_s})\right] + \mathrm{E}_{O_r}\left[\log(1 - D_1(G_{r\to s}(\boldsymbol{O_r})))\right] \\ L_{\text{adv}}^{O_r} &= \mathrm{E}_{O_r}\left[\log D_2(\boldsymbol{O_r})\right] + \mathrm{E}_{O_s}\left[\log(1 - D_2(G_{s\to r}(\boldsymbol{O_s})))\right] \end{aligned} \tag{1}$$

where $L_{\text{adv}}^{O_s}$ and $L_{\text{adv}}^{O_r}$ represent the adversarial loss constraints of the simulated image and the real image branches, respectively. D_1 and D_2 are two discriminators, $D_1(\boldsymbol{O_s})$ indicates the discrimination of the input simulated image, and $D_2(\boldsymbol{O_r})$ indicates the discrimination of the input real image. $G_{r\to s}(\boldsymbol{O_r})$ and $G_{s\to r}(\boldsymbol{O_s})$ stand for the process of generating images $\boldsymbol{B_r} - \boldsymbol{N_s}$ and $\boldsymbol{B_s} - \boldsymbol{N_r}$ in the first stage, where $G_{r\to s}$ includes G_3 and G_4, and $G_{s\to r}$ includes G_1 and G_2. Hence, $D_1(G_{r\to s}(\boldsymbol{O_r}))$ and $D_2(G_{s\to r}(\boldsymbol{O_s}))$ are the discrimination of $\boldsymbol{B_r} - \boldsymbol{N_s}$ and $\boldsymbol{B_s} - \boldsymbol{N_r}$, respectively.

2) MSE Loss: Utilizing the correspondence between ground truth backgrounds and simulated images, we can enforce supervised constraints within the simulated image branch. This approach leverages the fact that the simulated images are generated by introducing stripe noise to clean images, allowing us to establish a reliable basis for supervision and enhance the performance of the model. It can be expressed as

$$\begin{aligned} L_{\text{mse}} &= \|G_1(\boldsymbol{O_s}) - B\|^2 + \|G_1(G_{s\to r}(\boldsymbol{O_s})) - \boldsymbol{B}\|^2 \\ &= \|\boldsymbol{B_s} - \boldsymbol{B}\|^2 + \|\boldsymbol{B_s'} - \boldsymbol{B}\|^2 \end{aligned} \tag{2}$$

where L_{mse} represents the supervised constraint of the simulated image branch. $G_1(\boldsymbol{O_s})$ stands for the stripe noise removal of the input simulated image $\boldsymbol{O_s}$ by the generator G_1 to obtain the background component $\boldsymbol{B_s}$. $\boldsymbol{B}$ is the ground truth. $G_1(G_{s\to r}(\boldsymbol{O_s}))$ is the generation process of the background component $\boldsymbol{B_s'}$ derived from $\boldsymbol{B_s} - \boldsymbol{N_r}$.

3) Self-Consistency Loss: Maintaining consistency is crucial in both the decoupling of images and the transformation generation process, particularly in enhancing the model's generative capabilities through self-supervision. One key aspect that requires consistency is the preservation of the clean background image, which enhances the model's focus on the generation of stripe noise and its conversion.

$$\begin{aligned} L_{\text{cyc}}^{B_s} &= \mathrm{E}_{O_s}[\|G_1(\boldsymbol{O_s}) - G_1(G_{s\to r}(\boldsymbol{O_s}))\|_1] \\ &= \mathrm{E}_{O_s}\left[\|\boldsymbol{B_s} - \boldsymbol{B_s'}\|_1\right] \\ L_{\text{cyc}}^{B_r} &= \mathrm{E}_{O_r}[\|G_3(\boldsymbol{O_r}) - G_3(G_{r\to s}(\boldsymbol{O_r}))\|_1] \\ &= \mathrm{E}_{O_r}\left[\|\boldsymbol{B_r} - \boldsymbol{B_r'}\|_1\right]. \end{aligned} \tag{3}$$

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The second aspect pertains to the inputs and outputs of the model, where both the simulated and real images should remain unchanged, which can be written as follows:

$$\begin{aligned} L_{\text{cyc}}^{O_s} &= \mathrm{E}_{O_s}[\|\boldsymbol{O_s} - F_{r\to s}(G_{s\to r}(\boldsymbol{O_s}))\|_1] \\ &= \mathrm{E}_{O_s}\left[\|\boldsymbol{O_s} - \boldsymbol{O_s'}\|_1\right] \\ L_{\text{cyc}}^{O_r} &= \mathrm{E}_{O_r}[\|\boldsymbol{O_r} - F_{s\to r}(G_{r\to s}(\boldsymbol{O_r}))\|_1] \\ &= \mathrm{E}_{O_r}\left[\|\boldsymbol{O_r} - \boldsymbol{O_r'}\|_1\right]. \end{aligned} \tag{4}$$

$L_{\text{cyc}}^{B_s}$ and $L_{\text{cyc}}^{B_r}$ represent the consistency constraints on background component in the simulated and real image branches, respectively. $L_{\text{cyc}}^{O_s}$ and $L_{\text{cyc}}^{O_r}$ are the consistency constraints on image information of two branches. $F_{r\to s}$ and $F_{s\to r}$ refer to the generated images $\boldsymbol{O_s'}$ and $\boldsymbol{O_r'}$ in the second stage for simulated and real images, respectively. The self-consistency loss ensures consistency throughout the process and guarantees that the model accurately preserves the original information present in the input and output images.

4) Full Objective: The comprehensive objective function, encompassing adversarial, self-consistency, and MSE losses, can be expressed as follows:

$$\begin{aligned} L = \lambda_{\text{adv}}\left(L_{\text{adv}}^{O_s} + L_{\text{adv}}^{O_r}\right) + \lambda_{\text{mse}} L_{\text{mse}} \\ + \lambda_{\text{cyc}}\left(L_{\text{cyc}}^{B_s} + L_{\text{cyc}}^{B_r} + L_{\text{cyc}}^{O_s} + L_{\text{cyc}}^{O_r}\right) \end{aligned} \tag{5}$$

where λ_{adv}, λ_{mse}, and λ_{cyc} are hyperparameters corresponding to the three constraints, which are employed to balance the three parts of the full objective loss.

Regarding the three constraints mentioned above, the unsupervised adversarial loss is able to motivate the noise transformation between the simulated and real branches and further enhance the model to learn the real noise distribution. Self-consistent loss constrains the background components and images throughout the training process and is able to reduce the information loss caused by strip noise removal. Supervised loss effectively utilizes the ground truth of the simulated branch to enhance the performance of the model.

Including a comprehensive objective function in our framework addresses the challenge of lacking paired real degraded images. To preserve content information and maintain the desired output domain, each decoupling and generation subnetwork in our framework is subjected to a minimum of one unsupervised adversarial loss and one self-supervised self-consistency loss. These loss functions play a crucial role in utilizing the clean background and stripe noise components, guiding the overall process, and ensuring the desired outcomes.

C. Network Architecture

Since our work focuses on designing image decoupling and stripe noise transformation strategies, this section will briefly describe the architecture of the subnetworks, including G_1, G_2, G_3, G_4, D_1, and D_2, while the implementation details are provided in the **Supplementary Material**.

G_1 and G_3: In the proposed method, the generative models G_1 and G_3 utilize the U-net structure [51] to generate the clean background component from degraded input images (simulated and real images). As depicted in Fig. 2, the input image undergoes an encoding process, which involves a sequence of downsampling operations to extract deep features and expand the receptive field of the network. Subsequently, it proceeds through a decoding process, consisting of a series of upsampling operations, to generate the clean background image layer that matches the input size. Within the network, the information flow is enhanced by concatenating features of the same size from the encoding and decoding processes.

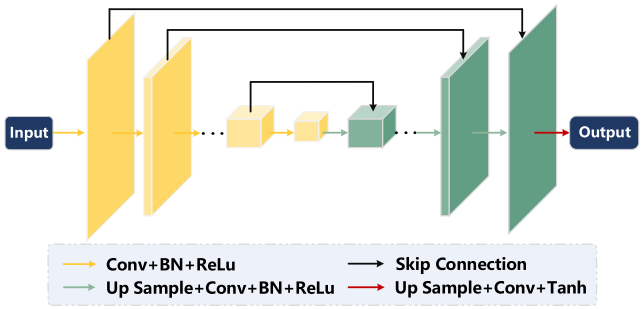


Fig. 2. Structure of the proposed G_1 and G_3, which uses U-net architecture. The process of generating the background components from the degraded image is performed by encoding and decoding operations.

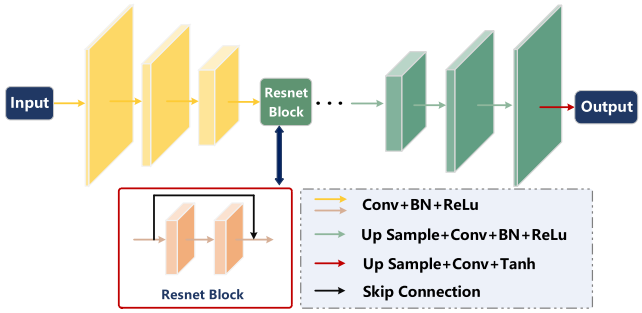


Fig. 3. Structure of the proposed G_2 and G_4, where the ResNet architecture is employed for transforming stripe noise between simulated and real domains.

G_2 and G_4: The G_2 and G_4 components of the generated model utilize the ResNet structure for intertransformation generation of the strip noise component between simulated and real data [52]. The network structure is shown in Fig. 3. The input degraded image goes through convolutional layers to expand the receptive field. Residual modules deepen the network without changing the input feature size. Finally, upsampling produces the output strip noise layer with the same dimensions as the original input. Feature abstraction is enhanced within the residual module by establishing connections between forward and backward feature information flow, improving the model's ability to extract meaningful features.

D_1 and D_2: In the proposed SDTNet, the discriminators D_1 and D_2 adopt the PatchGAN [53], [54] structure shown in Fig. 4, which differs from the traditional GAN discriminator. The traditional GAN discriminator consists of convolutional and fully connected layers, outputting a scalar value for overall discrimination. In contrast, PatchGAN utilizes an all-convolutional network structure, generating an $n \times n$ matrix where each element represents the discrimination of a local region of the input image (a specific region in the receptive field). The discrimination of the entire image is obtained by calculating the mean value of the matrix. This local focus

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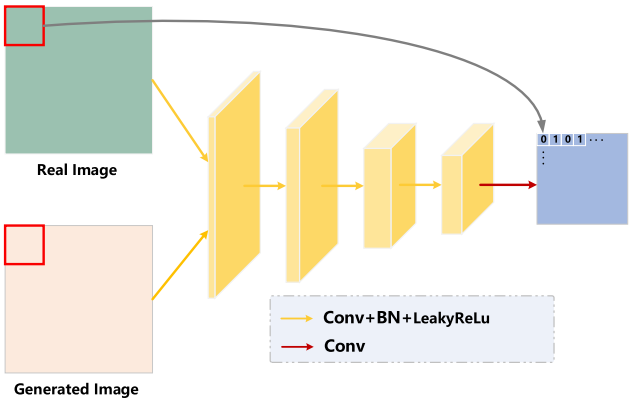


Fig. 4. Structure of the proposed D_1 and D_2. The PatchGAN architecture is utilized to discriminate the different images.

allows PatchGAN to capture fine-grained details and produce high-quality generated images.

D. Implementation Details

We cropped 256 × 256 image patches from the DFC2018 Houston data [55], obtaining 6000 samples by rotating them. The dataset was divided into training, validation, and test datasets in a 60%, 20%, and 20% split, respectively. For the real images, we used data from the full spectrum airborne hyperspectral imager (FAHI) [56], cropping the same size and number of samples as the simulated images. In the training of SDTNet, there are two main stages. In the first stage, the simulated data and real data are fed into their corresponding branches, G_1 and G_3, producing intermediate results, $\boldsymbol{B}_s - \boldsymbol{N}_r$ and $\boldsymbol{B}_r - \boldsymbol{N}_s$, through G_2 and G_4, respectively. In the second stage, these intermediate results are fed into G_3 and G_3, respectively, and processed again through G_4 and G_2, completing their respective branches. Finally, the overall loss is calculated to update the model parameters until the training is complete.

The gray values of the training images were normalized to the range [0, 1]. We empirically set the balance weight λ_{adv}, λ_{cyc}, and λ_{mse} as 2, 10, and 20. The Adam optimizer was employed during the model training process. Before training the SDTNet, the U-Net model was pretrained using the simulated data, with a learning rate of 0.00001 and an epoch set to 20. The trained parameters of the U-Net were then utilized to initialize G_1 and G_3 in the model. This initialization significantly contributes to expediting the training process of the entire model. The learning rate for G_1 and G_3 was set to 0.00001, while the learning rate for G_2, G_4, D_1, and D_2 was set to 0.0001. By assigning a smaller learning rate to the networks initialized with pretrained parameters and a larger learning rate to the others, the stability of the model training is promoted. The entire model was trained for 40 epochs, ensuring comprehensive convergence and performance improvement.

IV. Experiments

In this section, we conducted experiments to compare our proposed SDTNet with several classic destriping methods. These include MM [31], the anisotropic spectral–spatial TV (ASSTV) [11], the TV regularization and group sparsity model (TVGS) [57], the proximal alternating direction method of multipliers (PADMM) [58], as well as deep learning-based methods (ICSRN) [27], the spatial–spectral gradient network (SSGN) [28], column-spatial correction network (CSCNet) [50], and progressive recurrent network (PRNet) [23]. In addition, we also replace G_1 and G_3, in SDTNet with CSCNet and PRNet with exquisite network structure design to participate in the experiment comparison. They are recorded as SDTNet (CSCNet) and SDTNet (PRNet), respectively. The experiments were performed on multiple datasets. For the nondata-driven approaches (MM, ASSTV, TVGS, and PADMM), we conducted the experiments on a personal laptop equipped with MATLAB 2023b. The laptop had an Intel i5-10210U CPU with a clock speed of 1.6 GHz and 8 GB RAM. The data-driven methods (ICSRN and SSGN) were trained and tested on a workstation with the PyTorch 1.1.0 framework. The workstation featured an Intel i7-8700k CPU with a clock speed of 3.8 GHz, 128 GB RAM, and an NVIDIA GTX1080Ti GPU.

To assess the efficacy of SDTNet, we conducted a quantitative analysis to evaluate the performance of the models. This evaluation encompassed several metrics, including signal-to-noise ratio (SNR), peak SNR (PSNR), structural similarity index (SSIM), spectral angular distance (SAD), and inverse coefficient of variation (ICV). Additionally, we examined the nonuniformity of the restored images and conducted a mean cross-profile analysis to assess the effects of various methods on the destriping results.

A. Simulated Image Destriping

In this section, simulated data are generated by introducing additive noise to clean multispectral remote sensing data. DFC2018 Houston was used as a clean background in the experiment, which was made publicly available by the laboratory of Dr. Saurabh Prasad at the University of Houston and contains 48 bands of spectral data. The process of degraded data generation can be represented as follows:

$$V'_{l,j}(\varphi) = v_{l,j}(\varphi) + C_{l,j} \tag{6}$$

where $v_{l,j}(\varphi)$ stands for the value of pixel φ in the clean images. $C_{l,j}$ is set to a random value, but the same columns need to be set to the same value. Hence, we add strip noise to each column of the image and simulate the directional structure characteristic of strip noise in real situations. We retrained the data-driven models using identical training data and protocols to ensure a fair comparison. During the training and testing phases, we referenced the original implementations of these models to maintain consistency.

1) Qualitative Evaluation: Figs. 5 and 6 depict the visual results of stripe removal for all methods applied to the simulated image. It is evident that the deep learning-based methods outperform others in effectively eliminating noise from the simulated stripe noise. However, as shown in Fig. 6(c), the statistical-based MM method still exhibits slight noise artifacts even after stripe removal, particularly when confronted with significant material differences between columns. Similarly, optimization-based methods, as shown in Fig. 5(d)–(f) for ASSTV and PADMM, display local in-homogeneity after

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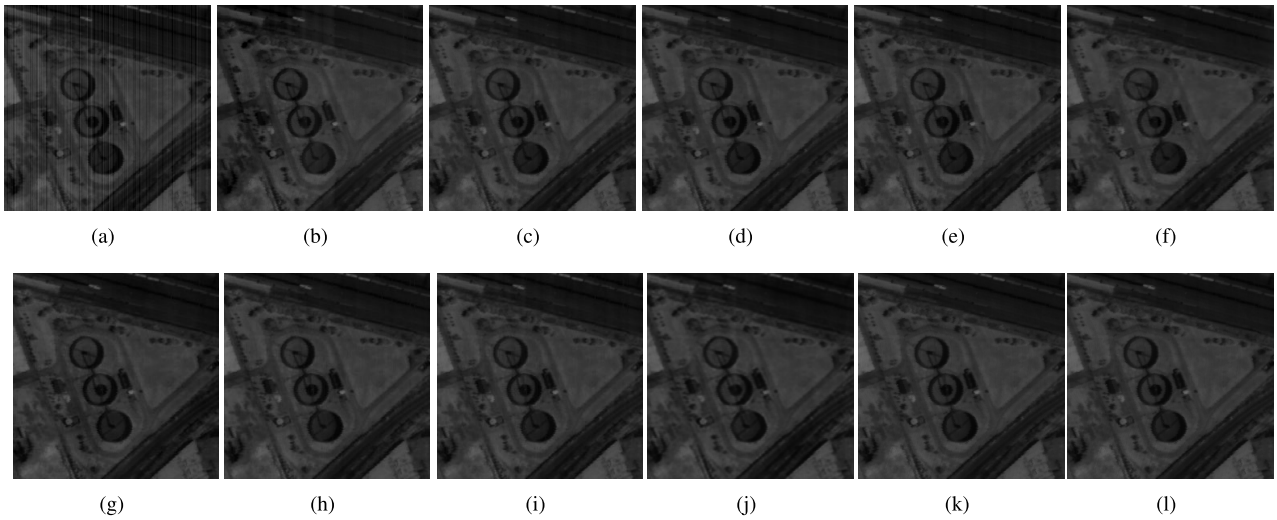


Fig. 5. Correction results of simulated image patch 1. (a) Simulated image. Image corrected by (b) MM, (c) ASSTV, (d) TVGS, (e) PADMM, (f) ICSRN, (g) SSGN, (h) CSCNet, (i) PRNet, (j) SDTNet, (k) SDTNet (CSCNet), and (l) SCTNet (PRNet).

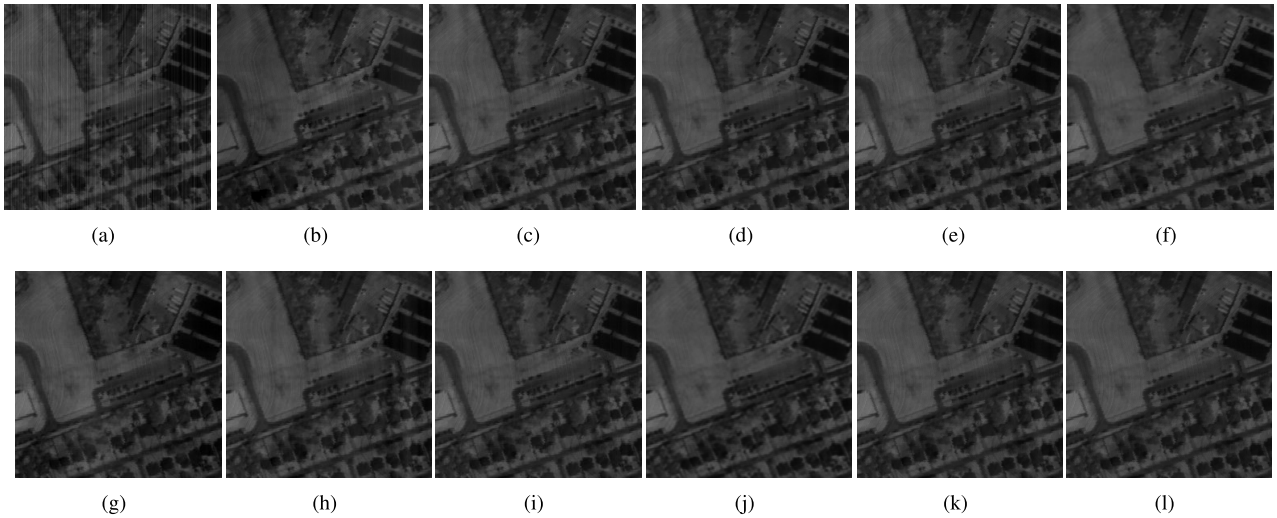


Fig. 6. Correction results of simulated image patch 2. (a) Simulated image. Image corrected by (b) MM, (c) ASSTV, (d) TVGS, (e) PADMM, (f) ICSRN, (g) SSGN, (h) CSCNet, (i) PRNet, (j) SDTNet, (k) SDTNet (CSCNet), and (l) SCTNet (PRNet).

destriping. The deep learning method performs better. However, we can see from Fig. 5(i) that PRNet does not completely remove the stripe noise, and residual noise can be seen in the upper right of the image. As shown in Fig. 5(l), SDTNet (PRNet) produces satisfactory results, verifying the effectiveness of the proposed image decoupling and noise transformation.

2) Quantitative Evaluation: In addition, this section provides a comprehensive quantitative evaluation of all comparison methods, utilizing four widely used evaluation indicators: SNR, PSNR, SSIM, and SAD. These metrics are commonly employed to assess the quality and similarity of reconstructed images. The results of the evaluation are presented in Table I, which offers a comparative analysis of the different methods. From Table I, it can be seen that SDTNet performs relatively well on the simulated dataset compared to simpler networks such as ICSRN and SSGN. This suggests that SDTNet is highly adept at reconstructing images with high fidelity and preserving the structural integrity of the original scene.

Furthermore, when considering the metrics of SNR and PSNR, SDTNet exhibits only a slight decrease in performance compared to the method known as ICSRN. However, this marginal difference signifies that the clean images produced by the proposed method in this chapter possess a higher overall quality. These results underscore the effectiveness of SDTNet in reducing noise and enhancing image clarity, thereby improving the perceptual quality of the reconstructed images. It is worth noting that SDTNet not only excels in terms of quantitative metrics but also demonstrates remarkable preservation of spatial and spectral information. Since CSCNet and PRNet possess complex network structure design, while SDTNet only adopts a simple U-Net structure, its evaluation metrics are not as good as the complex networks in simulated images. However, from the results of SDTNet (CSCNet) and SDTNet (PRNet), the image decoupling and noise transformation architecture using SDTNet can further promote the performance of the model.

B. Real Image Destriping

In order to evaluate the suitability of different methods, we conducted further investigations on the stripe removal results using real data. The real data utilized in the study

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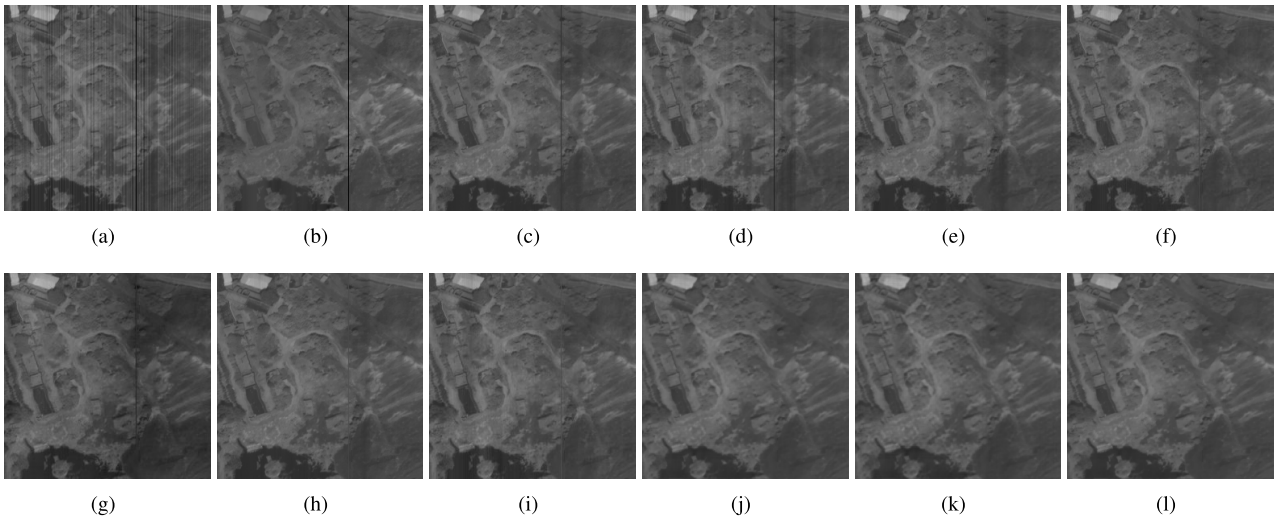


Fig. 7. Correction results of real image patch 1. (a) Real image. Image corrected by (b) MM, (c) ASSTV, (d) TVGS, (e) PADMM, (f) ICSRN, (g) SSGN, (h) CSCNet, (i) PRNet, (j) SDTNet, (k) SDTNet (CSCNet), and (l) SCTNet (PRNet).

TABLE I
FOUR INDICATORS OF DIFFERENT METHODS ON SIMULATED DATASETS

Method	SNR	PSNR	SSIM	SAD
MM	11.6872	26.4043	0.9142	0.1468
ASSTV	19.7034	33.4395	0.9710	0.0937
TVGS	20.3157	34.9452	0.9766	0.0963
PADMM	20.1352	34.3305	0.9718	0.0993
ICSRN	23.0660	37.2479	0.9832	0.0812
SSGN	22.1764	36.4925	0.9687	0.0884
SDTNet	23.1295	37.0665	0.9807	0.0805
CSCNet	30.0873	43.5218	0.9899	0.0531
PRNet	31.6538	45.7823	0.9915	0.0492
SDTNet(CSCNet)	31.0063	44.5536	0.9873	0.0489
SDTNet(PRNet)	31.9524	46.9648	0.9923	0.0435

were collected by the FAHI. Figs. 7–9(a) present the original degraded image, where Fig. 9 is a shortwave-infrared image (SWIR).

1) Qualitative Evaluation: The visualization of the destriping results of all the methods on real images is presented in Figs. 7 and 8(b)–(h). The results indicate that the proposed SDTNet exhibits exceptional performance in effectively eliminating stripe noise from real images, especially in scenarios involving intricate stripe noise distributions. Notably, the proposed method effectively restores image information in the presence of deadlines. This is evident in Figs. 7 and 8(b), where the column containing the deadline lacks any image information, rendering the statistical-based method incapable of recovering the corresponding information. While optimization-based methods partially recover image information associated with the deadline, as demonstrated in Figs. 7(c)–(e) and 8(d) and (e), the resulting images remain visually inconsistent due to the complex nature of the stripe noise. Similarly, the deep learning-based methods, ICSRN and SSGN, are unable to restore the missing image information caused by the deadlines. This limitation arises from the fact that their training samples do not encompass the stripe noise distribution found in real samples. From Figs. 7–9(h)–(l), it can be observed that even CSCNet and PRNet, which are outstanding in simulated images, are unable to deal with real images with the complex distribution of stripe noise. The effectiveness of SDTNet in dealing with real noise through image decoupling and noise transformation is confirmed.

TABLE II
ICV OF THE REAL IMAGE USING DIFFERENT COMPARISON METHODS

Method	Original	MM	ASSTV	TVGS
ICV	9.2415	48.6140	39.1768	29.2475
Method	PADMM	ICSRN	SSGN	CSCNet
ICV	14.1555	26.4968	18.6157	35.5890
Method	PRNet	SDTNet	SDTNet(CSCNet)	SDTNet(PRNet)
ICV	30.4607	36.8569	40.0125	38.7421

2) Quantitative Evaluation: We evaluate the recovery performance of each method on real data using ICV [59], defined as

$$\mathrm{ICV} = \frac{R_m}{R_s} \tag{7}$$

where R_m and R_s are the mean and standard deviation of pixel values. The ICV index is computed in homogeneous regions within a window of 20 $\times$ 50 pixels, as shown in Fig. 10. Table II gives the calculation results of the different methods.

From Table II, it can be seen that the two methods, MM and SDTNet (CSCNet), have the highest ICV values. The proposed SDTNet also effectively improves the uniformity of homogeneous regions. It is worth noting that although the MM has the highest ICV value, it does not restore the image, as can be seen from Fig. 9(b) that its image quality after destriping is poor. Similarly, ASSTV also has high ICV value but low restoration quality.

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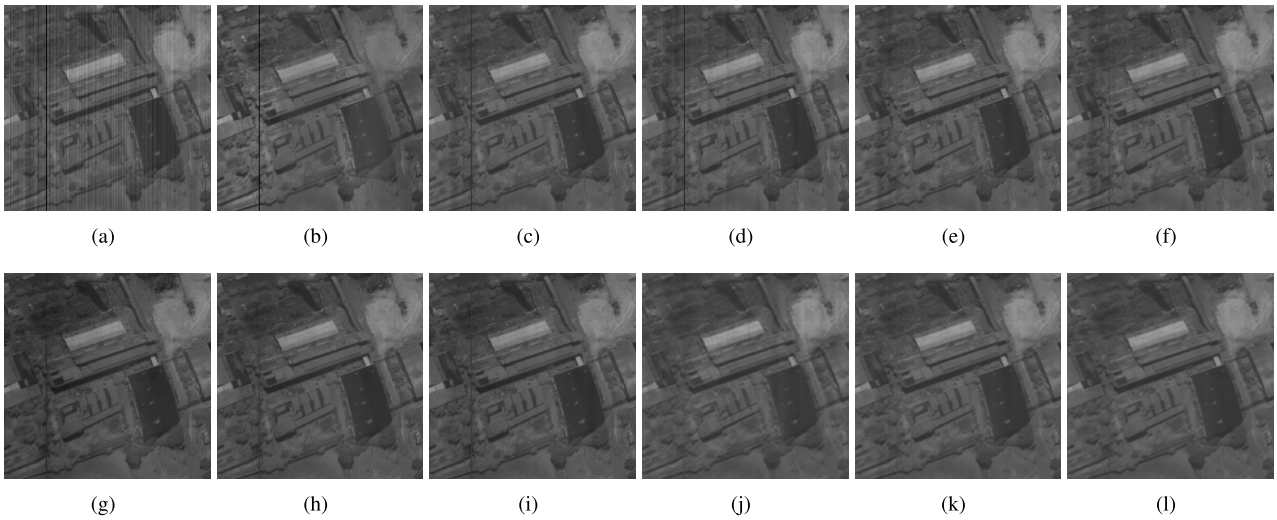


Fig. 8. Correction results of real image patch 2. (a) Real image. Image corrected by (b) MM, (c) ASSTV, (d) TVGS, (e) PADMM, (f) ICSRN, (g) SSGN, (h) CSCNet, (i) PRNet, (j) SDTNet, (k) SDTNet (CSCNet), and (l) SCTNet (PRNet).

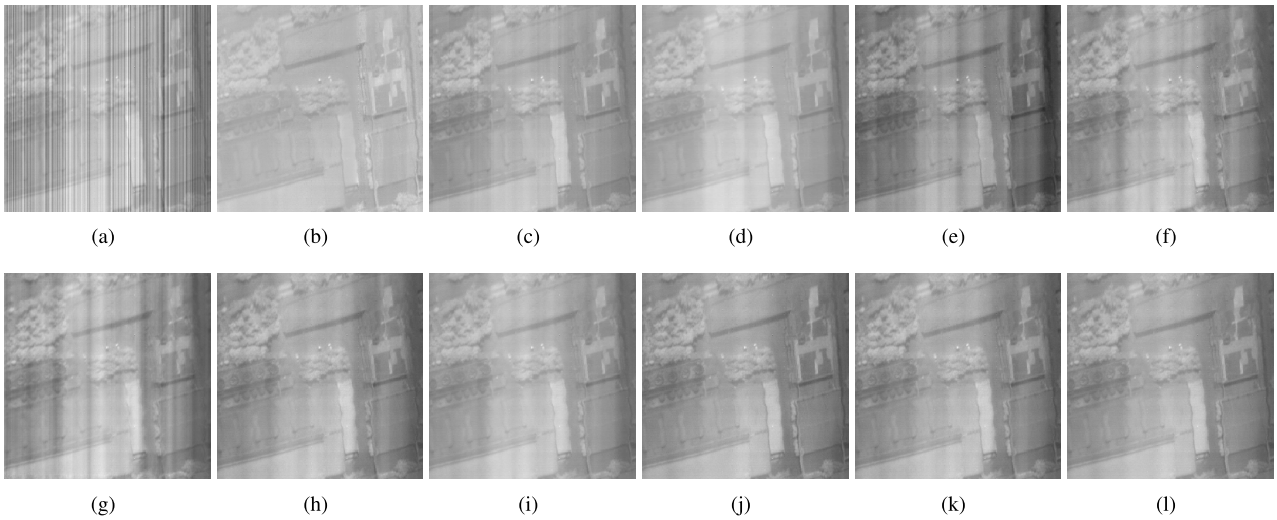


Fig. 9. Correction results of real image patch 3. (a) Real image. Image corrected by (b) MM, (c) ASSTV, (d) TVGS, (e) PADMM, (f) ICSRN, (g) SSGN, (h) CSCNet, (i) PRNet, (j) SDTNet, (k) SDTNet (CSCNet), and (l) SCTNet (PRNet).

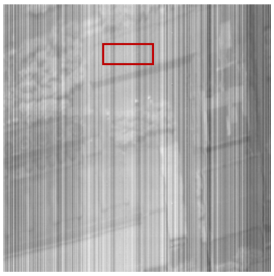


Fig. 10. Region used to calculate the ICV.

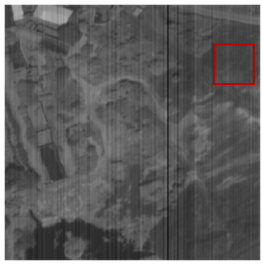


Fig. 11. Ground target region.

C. Image Uniformity

Destriping aims to enhance the uniformity of degraded images. Accordingly, the effectiveness of destriping can be quantitatively assessed by analyzing the degree of nonuniformity in the restored images. It can be written as follows:

$$U = \sqrt{\frac{\sum_{i=1}^{N_s} \left(S_i - \bar{S}\right)^2}{N_s}}. \tag{8}$$

Let U denote the measure of nonuniformity in the image. N_s represents the total number of samples, S_i denotes the value of pixel i, and $\bar{S}$ represents the mean value of all S_i pixels. Reduced nonuniformity values indicate a significant improvement in the uniformity of the restored image. To evaluate this, we selected a uniform region of size 40×40 from the real images, indicated by the red box in Fig. 11. The computation results for all methods are presented in Table III.

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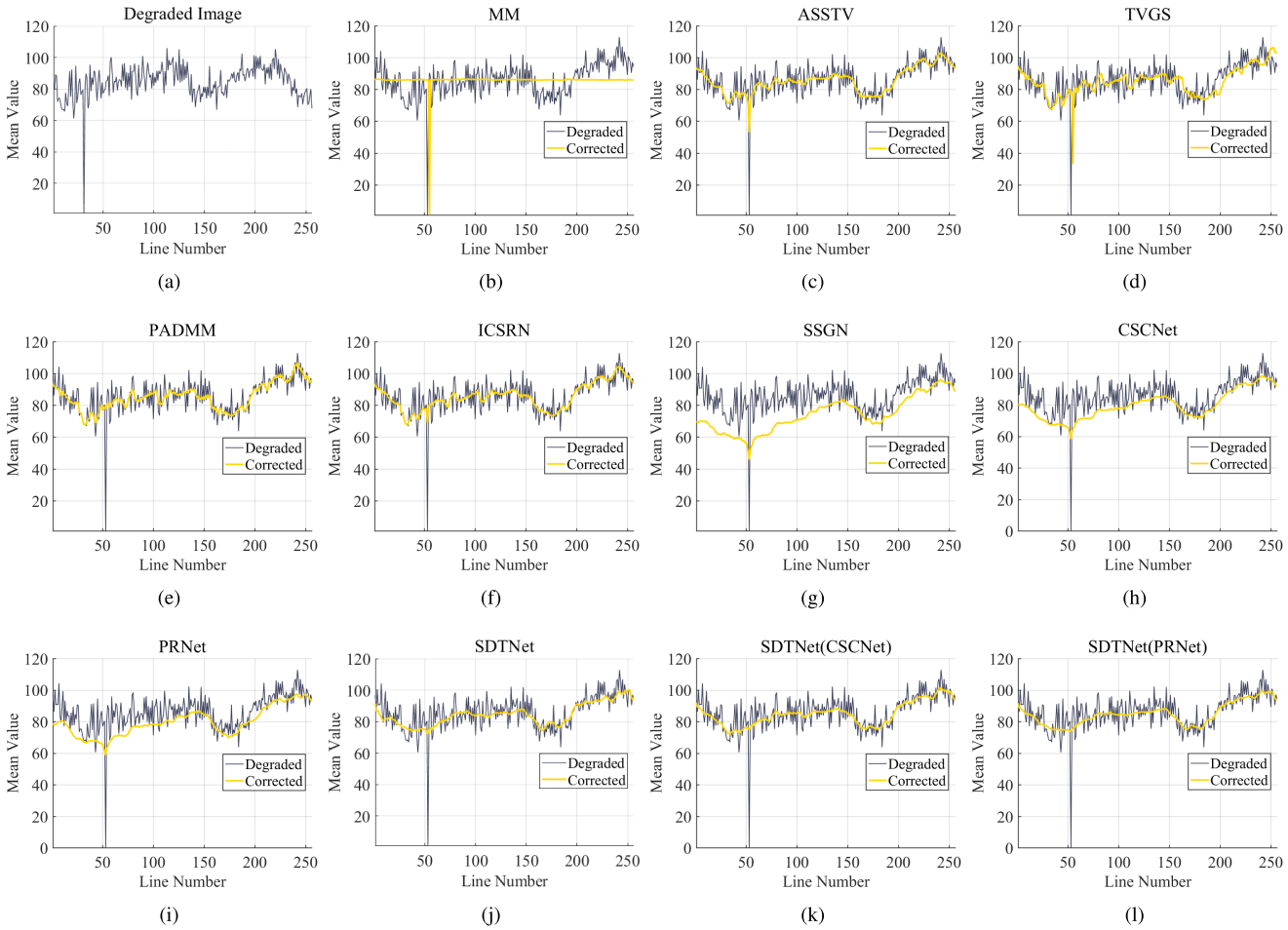


Fig. 12. Mean cross-profile curves of real image patch 2. (a) Real image. Image corrected by (b) MM, (c) ASSTV, (d) TVGS, (e) PADMM, (f) ICSRN, (g) SSGN, (h) CSCNet, (i) PRNet, (j) SDTNet, (k) SDTNet (CSCNet), and (l) SCTNet (PRNet).

TABLE III
IMAGE NONUNIFORMITY USING DIFFERENT COMPARISON METHODS

Method	Original	MM	ASSTV	TVGS
U	0.0731	0.0769	0.0558	0.0631
Method	PADMM	ICSRN	SSGN	CSCNet
U	0.0657	0.0551	0.0821	0.0566
Method	PRNet	SDTNet	SDTNet(CSCNet)	SDTNet(PRNet)
U	0.0573	0.0564	0.0594	0.0586

From Table III, it can be observed that the proposed SDTNet exhibits a slightly higher nonuniformity value compared to ASSTV and ICSRN, indicating a relatively small level of nonuniformity. Fig. 7(c) reveals the presence of residual stripe noise in the images restored by ASSTV, despite having lower local nonuniformity values. Therefore, the overall correction effect of ASSTV is not as effective as SDTNet. Additionally, as shown in Fig. 7(f) and (h), although the nonuniformity values of the proposed method are slightly higher than those of ICSRN, SDTNet performs better in terms of restoring the deadlines. In summary, SDTNet effectively removes stripe noise and improves the overall image uniformity.

D. Mean Cross-Profile Analysis

Fig. 12 presents the average cross section analysis of the recovery results, allowing for a more intuitive assessment of the band removal effect [29]. The horizontal coordinate represents the row number (column), while the vertical coordinate represents the average value of all pixels in the corresponding column. In particular, the average cross section profile of the corrected image should exhibit smoother characteristics, as depicted in Fig. 12.

Fig. 12 displays the mean cross-profile curves of the real image, indicating noticeable interference from stripe noise, particularly affected by the deadlines. Subsequently, Fig. 12(b)–(h) presents the mean cross-profile curves of the corrected image. It can be observed that all the destriping methods alleviate the noise interference to some extent. However, certain methods, such as MM, ASSTV, and TVGS, fail to remove the stripe noise, specifically the deadlines completely. Among these methods, SSGN results in a more significant alteration of the image information after band removal [see Fig. 12(g)]. Conversely, PADMM, ICSRN, and SDTNet methods successfully repair the deadlines, with the mean cross-profile curves appearing relatively smooth following SDTNet correction. Because of that, the mean cross-profile curves serve as additional confirmation of the effectiveness of our proposed method. Fig. 12 also further validates the

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TABLE IV
FOUR INDICATORS FOR SDTNET USING DIFFERENT AMOUNTS OF TRAINING DATA ON SIMULATED DATASETS

Training Sample Size	SNR	PSNR	SSIM	SAD
40% (2400)	20.8973	34.7649	0.9683	0.0926
60% (3600)	23.1295	37.0665	0.9807	0.0805
80% (4800)	23.0782	37.5317	0.9828	0.0824

TABLE V
NETWORK PARAMETERS OF FIVE DEEP LEARNING-BASED MODELS

Method	ICSRN	SSGN	CSCNet	PRNet	SDTNet
Parameters(M)	0.8	0.2	6.2	0.39	13

ability of SDTNet to handle real data. The advantages of the supervised methods in simulated the data can be seen in Section IV-A, such as the satisfactory results of CSCNet and PRNet on the four evaluation metrics. However, as shown in Fig. 12(h) and (i), their results for real images are poor. In contrast, Fig. 12(k) and (l) confirms that the image decoupling and noise transformation approach designed by SDTNet can effectively promote the model's learning of real noise and enhance the restoration results.

E. Training Samples Analysis

We investigated the image of the size of the training set on the performance of the model. In addition, 40%, 60%, and 80% of the total samples are taken to train the model, and the corresponding experimental results are shown in Table IV. We can observe that the model's performance improves as the number of training samples increases. When 40% of the samples are used to train the model, the lack of training samples leads to the poorer results of SDTNet. Since the network design is simpler and the amount of data is sufficient to train the model, 60% and 80% of the training samples correspond to similar performance of the model.

F. Model Complexity Analysis

In order to conduct a comprehensive evaluation of the destriping models, we analyzed their respective tradeoffs between effectiveness and efficiency. Among them, the network parameters of the five deep learning-based methods and the inference time of each method are listed in Tables V and VI. From Table V, it can be seen that ICSRN, SSGN, and PRNet are all lightweight models. Due to the inclusion of six subnetworks in the SDTNet model, the model parameters are relatively large. However, during testing, we only use the U-Net structured generative network to remove stripe noise. Thus, the inference time of SDTNet is quite fast, second only to MM and SSGN. This discrepancy can be attributed to the simpler design and model structures of MM and SSGN, which allow for faster processing. As observed in the table, it is evident that the MM method and deep learning-based methods exhibit faster image processing speeds compared to other approaches. Nevertheless, when considering the previous experimental results, it is evident that SDTNet achieves superior destriping performance while maintaining a relatively low inference time.

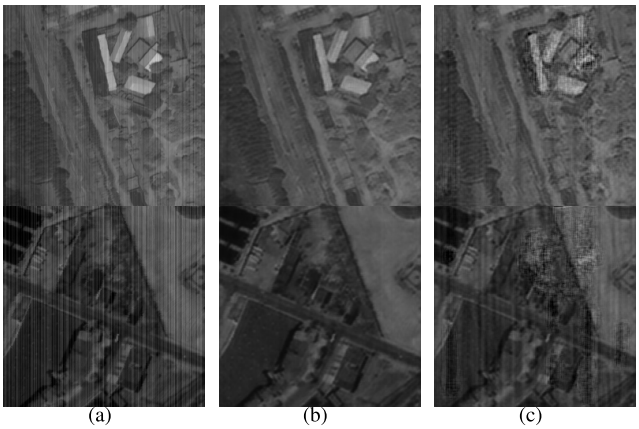


Fig. 13. (a) Original image. (b) Corrected images obtained by SDTNet. (c) Images obtained by the image transformation strategy.

G. Ablation Study

1) Effectiveness of the Stripe Noise Transformation: In the proposed SDTNet, G_2 and G_4 are specifically designed for stripe noise transformation between different domains, enabling the model to learn the distribution of real stripe noise. Our approach is referred to as a stripe transformation strategy, which focuses on relatively simple tasks to facilitate the training process of the model. To assess the effectiveness of this strategy, we directly input B_s and B_r to G_2 and G_4, respectively, resulting in $B_s - N_r$ and $B_r - N_s$ generation. This methodology is denoted as the image transformation strategy, as depicted in Fig. 13(c).

From Fig. 13, it is evident that the image transformation yields inferior results, with a significant loss of image detail information. Furthermore, blurring occurs in regions with dense feature targets. Consequently, training the model successfully using the image transformation strategy becomes challenging. In contrast, the destriping results of SDTNet, shown in Fig. 13(b), demonstrate that adopting the stripe noise transformation strategy effectively promotes the training success of the model.

2) Analysis of Hyperparametric Weighting Coefficients: In the loss function of SDTNet, we incorporate adversarial, self-consistency, and MSE constraints for model optimization and assign weighting coefficients to them, respectively. Therefore, it is essential to analyze the impact of different weighting coefficients on the model performance. Since these coefficients are set to adjust the three parts of the loss to similar magnitudes, we conduct experiments on simulated images with three levels for λ_{cyc} and λ_{mse} $(-10, 1, 10)$, respectively, when $\lambda_{\text{adv}} = 1$. We observe the effect of the weighting coefficients on the image restoration results, as presented in Table VII. We observe that the combination with $\lambda_{\text{adv}} = 1$, $\lambda_{\text{cyc}} = 10$, and $\lambda_{\text{mse}} = 10$ performs the best. Compared to λ_{mse}, λ_{cyc} has a greater impact on the overall restoration quality of the image, with both PSNR and SSIM decreasing as λ_{cyc} decreases. As MSE is a supervised constraint on the simulated branch, it is more crucial for the spatial structural information of the

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TABLE VI
INFERENCE TIME (IN SECONDS) OF EVERY METHOD ON THE IMAGE WITH THE SIZE OF 256 × 256 (FOR ONE BAND)

Method	MM	ASSTV	TVGS	PADMM	ICSRN	SSGN	CSCNet	PRNet	SDTNet
Time	0.01	8.77	5.27	3.05	0.26	0.07	0.61	0.17	0.13

TABLE VII
COMPARISON OF PSNR AND SSIM ON SIMULATED DATA WITH DIFFERENT HYPERPARAMETRIC WEIGHTS FOR $\lambda_{\text{ADV}} = 1$

λ_{cyc}	λ_{mse}	PSNR	SSIM
-10	-10	29.6532	0.9567
-10	1	29.7858	0.9543
-10	10	30.8021	0.9602
1	-10	33.1074	0.9596
1	10	33.6547	0.9685
10	-10	34.9853	0.9618
10	1	35.2362	0.9735
10	10	36.7856	0.9763

TABLE VIII
ABLATION STUDY FOR DIFFERENT TRAINING STRATEGIES ON THE SIMULATED DATASET. TRAINING 1 AND TRAINING 2 DENOTE TWO DIFFERENT TRAINING STRATEGIES

Strategy	PSNR	SSIM
Training-1	34.8769	0.9627
Training-2	36.1523	0.9726
Proposed	37.0665	0.9807

image. Setting a different λ_{mse} has a greater impact on the SSIM metrics of the image.

3) Effectiveness of Training Strategy: In Table VIII, we assess the effectiveness of various training strategies. Training 1 entails training the subnetworks in SDTNet without pretraining. Training 2 involves pretraining on a paired simulated dataset to remove subnetworks G_1 and G_3, followed by joint training with the learning rate divided by 100. The quantitative results highlight the importance of initializing subnetwork parameters through pretraining.

V. CONCLUSION

To tackle the challenge of inconsistent noise distribution between simulated training samples and real test samples faced by deep learning models, we proposed the SDTNet that effectively combines the decoupling of the degraded image and transformation generation of the stripe noise component between the simulated and real images. The model incorporates the real stripe noise distribution into the training process, improving its adaptation to real multispectral images affected by stripe noise. Extensive experiments on simulated and real data were conducted to validate the effectiveness of the proposed model. The comprehensive qualitative and quantitative evaluations demonstrated that SDTNet successfully removes image stripe noise, improves image homogeneity, and preserves the structural information of the original image. Moreover, it outperforms other methods when applied to real images, highlighting its superior adaptability. Overall, our work contributes to advancing multispectral image destriping by addressing the challenge of inconsistent noise distribution and achieving significant improvements in image quality and preservation of important image features. In the real image branch of the proposed method, we rely solely on unsupervised constraints for learning. To improve the model's understanding of real data, future research can explore the integration of additional techniques such as self-supervised learning and incorporation of prior knowledge.

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